Hardware

iGEM ·WW2

1. Overview

This project addresses the need for intervention based on the grading of necrosis severity in small cell lung cancer (SCLC) and designs a handheld SCLC grading intelligent nebulization terminal, serving as the hardware platform for the LDH detection combined with nebulization intervention scheme. The device utilizes the ESP32 as the core controller and integrates functions such as dual-channel temperature control, three-channel precise pump control, piezoelectric nebulization, and human-computer interaction. It fulfills the core requirements of individual temperature control and viability maintenance for the engineered bacterial chamber, as well as sequential spraying of bacteria according to LDH grading and proportional drug spraying. Simultaneously, the structure and hardware design maximize the protection of the activity of engineered bacteria and the spatial structure of protein drugs (IFN-γ).

Traditional medical nebulizers only achieve simple atomization of liquid medicine, lacking graded drug delivery and live bacteria preservation design, and cannot meet the personalized intervention needs of SCLC. This device focuses on precise atomization function, automatically executing a preset atomization scheme by inputting LDH grading results through buttons. In terms of hardware, it adopts vibrating mesh nebulization technology to reduce shear force, and a dual-chamber heat insulation design to achieve differentiated environmental control. In terms of software, it utilizes MicroPython to achieve intelligent scheduling of temperature control closed-loop and atomization process.

2. Device Design

2.1 Overall structure and composition

The device consists of the following core modules：

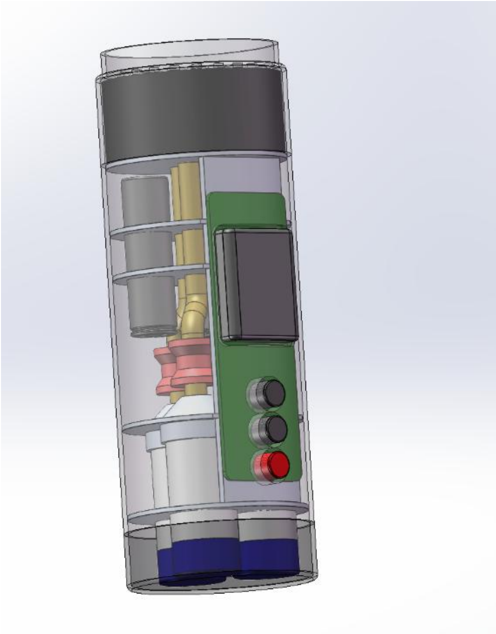
Module	Size	Function
Master Control Module	ESP32-WROOM-32E	The main controller is responsible for temperature control algorithms, pump speed adjustment, and atomization processes
Temperature sensing module	DS18B20 × 3	Detect temperature
pump control module	mini peristaltic pump × 3 +	Drug delivery, precise

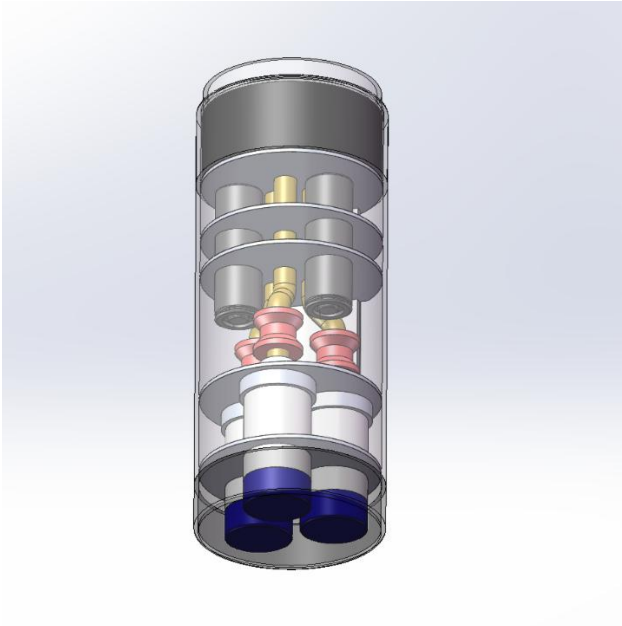
Module	Size	Function
	DRV8833 × 2	speed regulation
heating module	AO3400 MOS	control temperature
Atomization module	Vibrating screen mesh atomizing plate + atomizing drive board	Utilizing low shear force vibrating mesh technology, it generates an ideal pulmonary deposition particle size of 1-5μm, achieving liquid atomization
Power Supply	Lithium battery	Device power source
structural protection	Air insulation layer + 0.22μm PTFE filter membrane	The three-compartment chamber is equipped with a 1mm air insulation layer, and a top-mounted filter membrane to achieve pressure balance and biological safety protection, preventing bacterial fluid leakage
Display + interaction module	0.96-inch I2C OLED + 3x tactile buttons	OLED displays device status, temperature, and nebulization process, with 3 buttons corresponding to LDH classification levels I/II/III nebulization schemes

2.2 3D modeling

SolidWorks was used for parametric 3D modeling. The following parts and the full assembly were completed:

- Housing: Handheld housing, with sides cut flat to fit the screen/buttons
- Inner liner of the third chamber: 1mm insulation gap between chambers
- Y-shaped mixing flow channel, atomizing head assembly, PCB board, etc
- General drawing





3. Circuit Design

3.1 Circuit Overview

The circuit uses the ESP32 as the core microcontroller, with peripheral circuits built around it to fulfill core functionalities: The power supply end is connected to a TP4056 charging chip and an AMS1117-3.3V voltage regulator chip to achieve lithium battery charging and stable 3.3V power supply. The OLED display is connected via an I2C bus (SCL connected to GPIO22, SDA connected to GPIO21), and three hierarchical buttons are triggered by pull-up inputs connected to GND. Two DRV8833 chips drive three micro pumps, and an AO3400 MOSFET controls the switches of the heating plate and atomizing plate. Three DS18B20 temperature sensors are connected in parallel through GPIO4, with a series 4.7kΩ pull-up resistor to ensure signal stability, forming a complete temperature control, drive, and interaction circuit logic.

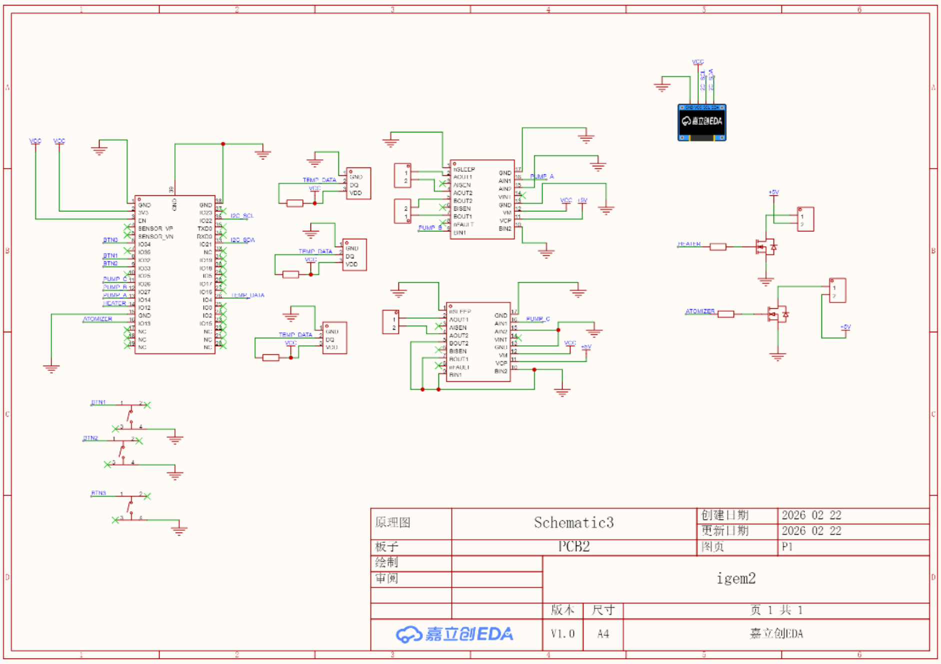
3.2 Pin Assignment

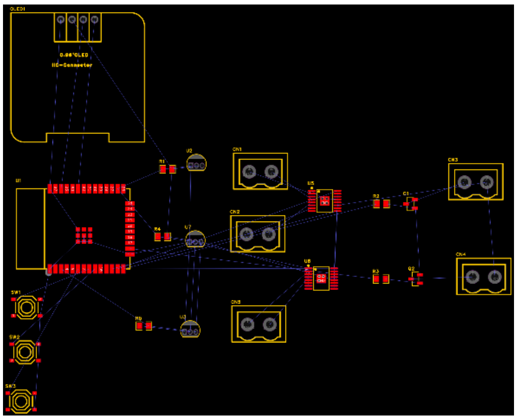
PIN	Function
IO4	Three-channel temperature sensor with single bus data port
IO12	On-off control of heating film in engineered bacteria storage
IO13	Power switch control for atomizing plate drive board
IO14	Engineered bacterial pump PWM speed regulation
IO27	Drug 1 pump PWM speed regulation
IO26	Drug 2 pump PWM speed regulation
IO32	LDH Grade I (mild necrosis) aerosolization trigger
IO33	LDH grade II (moderate necrosis) aerosolization trigger
IO34	LDH grade III (severe necrosis) triggered by nebulization
IO22	OLED screen I2C clock line
IO21	OLED screen I2C data cable
3.3V/GND	Power supply and ground

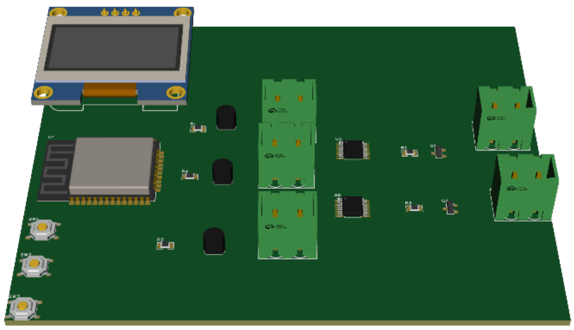
3.3 EDA Design Files

The complete circuit schematic and PCB layout have been designed using Lattice EDA, with the following design outputs

- Circuit schematic diagram: Shows the connection relationship between various components
- PCB layout: Printed circuit board wiring and pad design
- PCB 3D effect diagram: 3D rendering







4. Firmware & Algorithm

4.1 State Machine Workflow

The device firmware adopts a three-state closed-loop design of "standby - nebulization execution - end feedback", with the following workflow:

Standby → Atomization execution → End feedback → Standby (loop)

1. Standby mode: The green light is constantly on, and the OLED displays "SCLC Nebulizer Ready", with the current temperature of the engineered bacterial chamber being refreshed in real time (e.g., "Temp: 37.0℃"). It waits for the user to trigger nebulization via the leveling button.

2. Atomization execution status: Upon detecting the low-level trigger of the corresponding classification button, the OLED displays the current atomization phase ("Phase1: Bacteria" / "Phase2: Medicine"), remaining time, and real-time

temperature. The system follows the preset logic to execute the "spray bacteria first, then mix medicine" process, continuously monitoring temperature through DS18B20 and dynamically adjusting the heating status during this process.

3. End feedback status: After the nebulization process is completed, the OLED displays "Nebulization Done" and "Remove Mouthpiece"; if an abnormal temperature (>30℃) is detected in the medicine chamber, the red light stays on for 4 seconds, and the OLED displays "Temp Alarm! Check Medicine". After the fault is resolved, it automatically returns to the standby state.

4.2 core algorithm

1. Hierarchical atomization ratio algorithm

Set differentiated atomization parameters based on LDH classification to achieve precise ratio and timing control of engineered bacteria and dual drugs

2．Temperature closed-loop control algorithm

Using a simple on-off temperature control logic, the engineered bacterial storage chamber is maintained within the temperature range of 36.5℃~37.5℃

Control logic: When the DS18B20 detects a temperature <36.5℃, activate the heating film; when the temperature >37.5℃, deactivate the heating film; maintain the current heating status within the range to avoid temperature fluctuations caused by frequent start-stop.

Tips: Temperature, necrosis degree, and drug ratio are all assumed

5. Future Work（未来展望）

- Complete physical processing and debugging, verify atomization particle size, temperature control accuracy, and graded drug delivery accuracy, and optimize hardware reliability.
- Calibrate the atomization delivery efficiency based on biological experimental data, adjust the pump speed ratio and atomization duration, and align with biological intervention requirements.
- Expand the ESP32 WiFi/BLE wireless capabilities to enable data synchronization, remote adjustments, and storage of treatment records.
- Optimize the modularized medicine storage design, add a low liquid level alarm, and enhance the practicality and safety of the equipment.